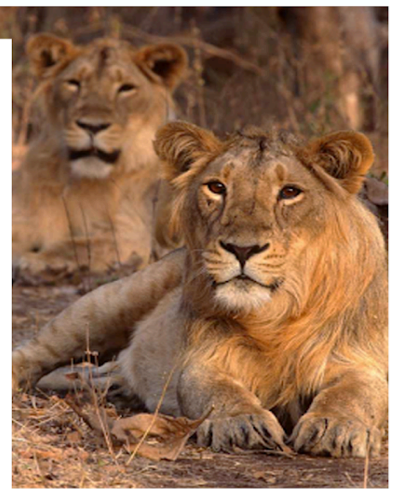
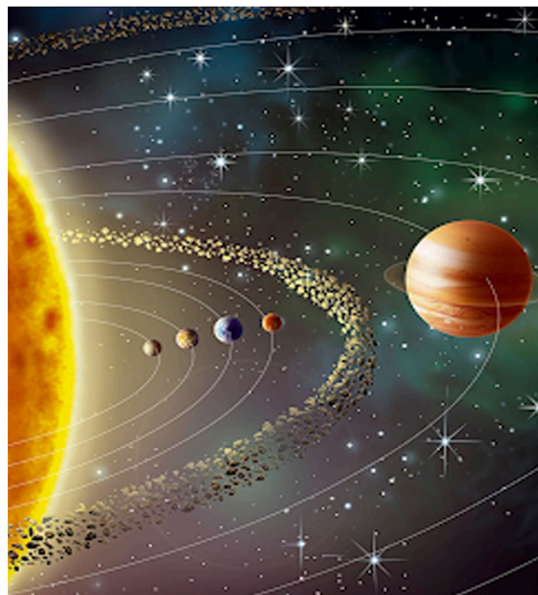
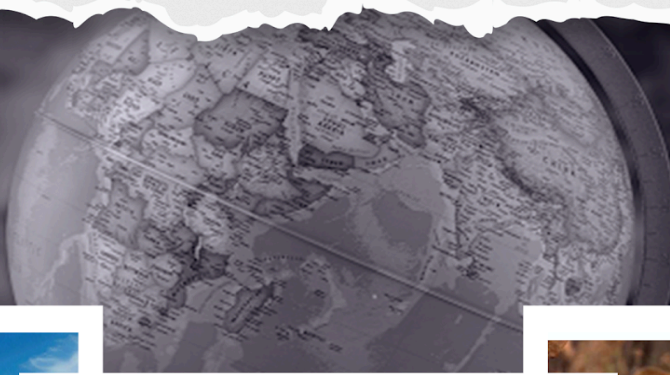


General Knowledge

World Geography

STUDY NOTES

Oceanography: Ocean Currents

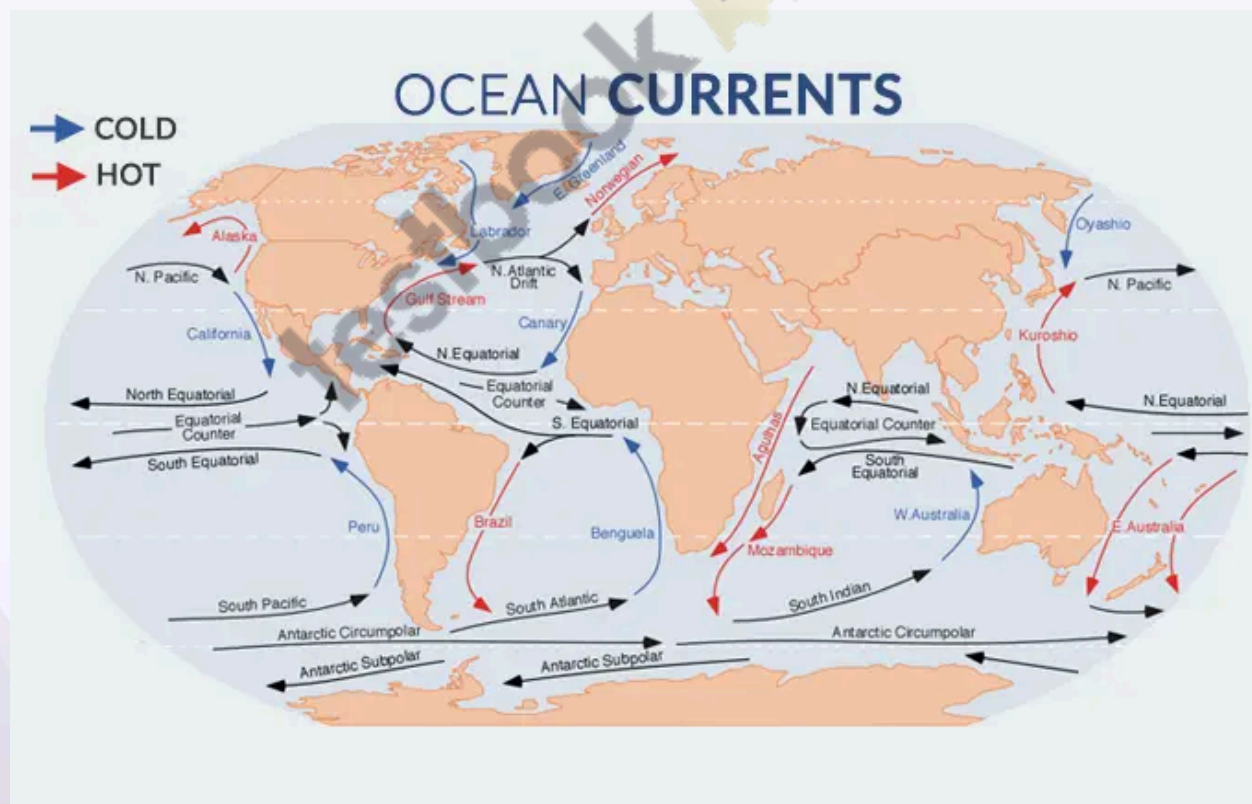


Oceanography: Ocean Currents

Ocean currents are large-scale movements of water within the world's oceans, driven by various forces, including wind, the Earth's rotation, temperature differences, salinity variations, and the gravitational pull of the moon and sun. Here are some key points about ocean currents.

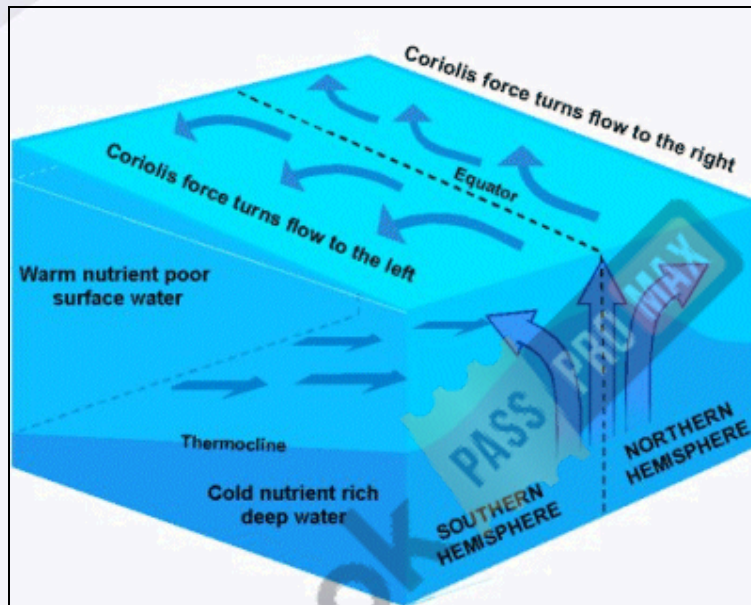
Types of Ocean Currents

- **Surface Currents:** These currents occur in the upper 400 meters of the ocean and are primarily driven by wind.
 - Examples include the Gulf Stream and the California Current.
- **Deep Water Currents:** Also known as thermohaline circulation, these currents are driven by differences in water density, which is affected by temperature and salinity.
 - They occur deeper in the ocean and play a crucial role in global climate.
- **Cold currents:** These currents are cooler than the surrounding water.
- **Warm currents:** These currents are often warmer than the surrounding water.



Causes of Ocean Currents

- **Wind:** Surface currents are primarily generated by wind patterns, which push water in specific directions.
- **Coriolis Effect:** The rotation of the Earth causes moving water to turn and twist, affecting the direction of currents. In the Northern Hemisphere, currents turn clockwise, while in the Southern Hemisphere, they turn counterclockwise.



- **Temperature and Salinity:** Variations in temperature (warm vs. cold water) and salinity (fresh vs. salt water) create density differences, driving deep-water currents.
- **Gravitational Forces:** The gravitational pull of the moon and the sun causes tides, which can also influence ocean currents.

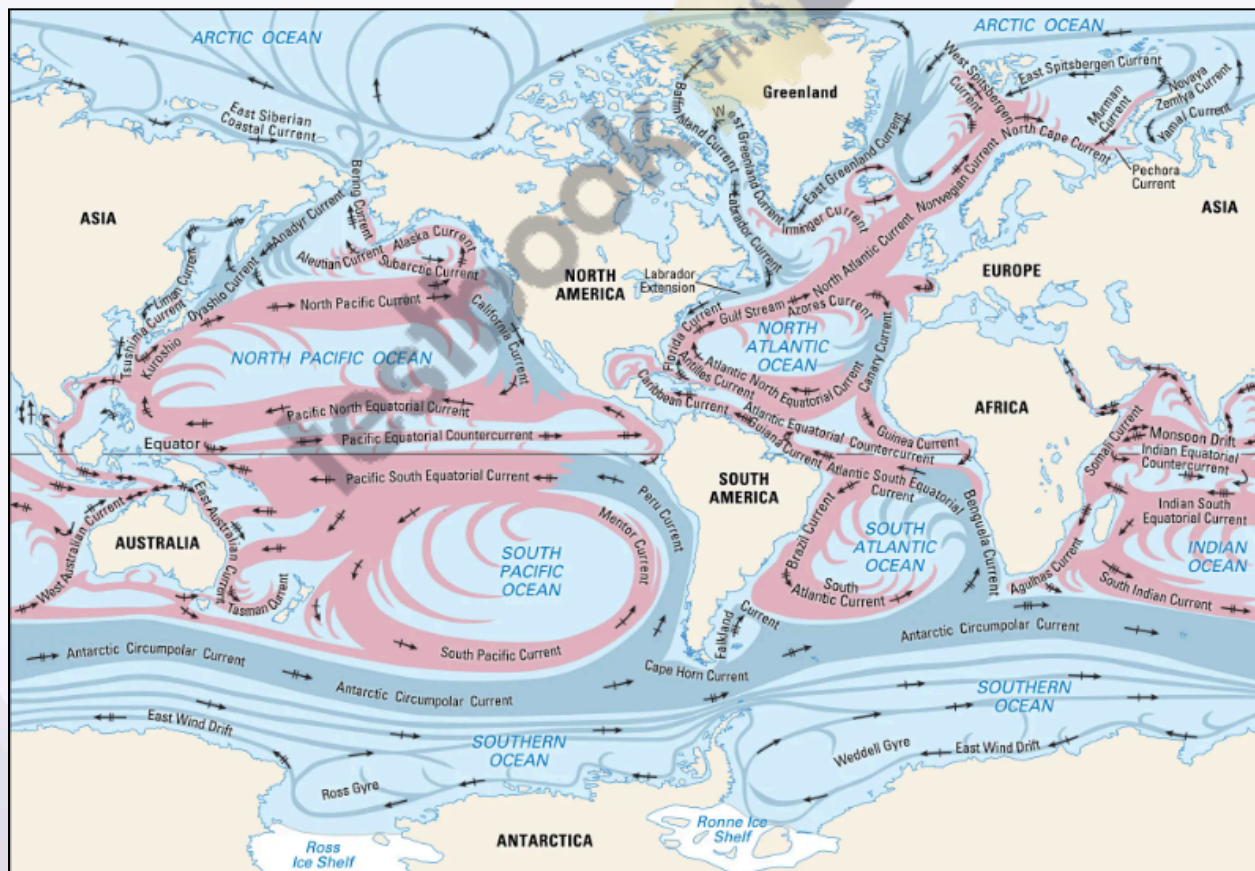
Importance of Ocean Currents

- **Climate Regulation:** Ocean currents play a critical role in regulating the Earth's climate by distributing heat around the planet. For example, the Gulf Stream transports warm water from the tropics to the North Atlantic, influencing weather patterns.
- **Marine Ecosystems:** Currents affect nutrient distribution, impacting marine life and ecosystems. They can carry nutrients from the ocean floor to the surface, supporting phytoplankton growth and the entire food web.
- **Navigation and Shipping:** Understanding ocean currents is vital for maritime navigation and shipping, as they can influence travel times and fuel efficiency.

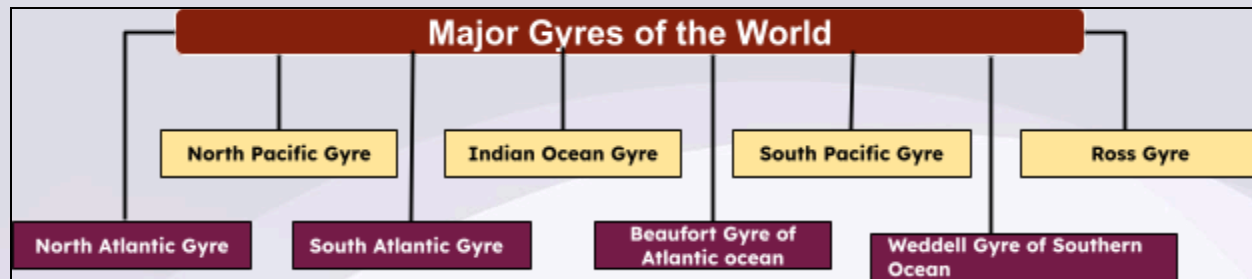
- **Weather Patterns:** Currents can influence weather patterns, including storm development and precipitation levels. El Niño and La Niña events, which are related to ocean currents in the Pacific, significantly affect global weather.

Gyres

- Gyres are large-scale circular ocean currents that are driven by global wind patterns and the Coriolis effect. They play a significant role in the movement of ocean waters and the distribution of heat and nutrients across the oceans. Here are some key points about gyres.
- Gyres are formed by the interaction of wind, the Earth's rotation (Coriolis effect), and the shape of the ocean basins.
- The Coriolis effect causes moving water to turn, leading to the circular motion characteristic of gyres.



Note: Red= Warm currents, Blue= Cold Currents



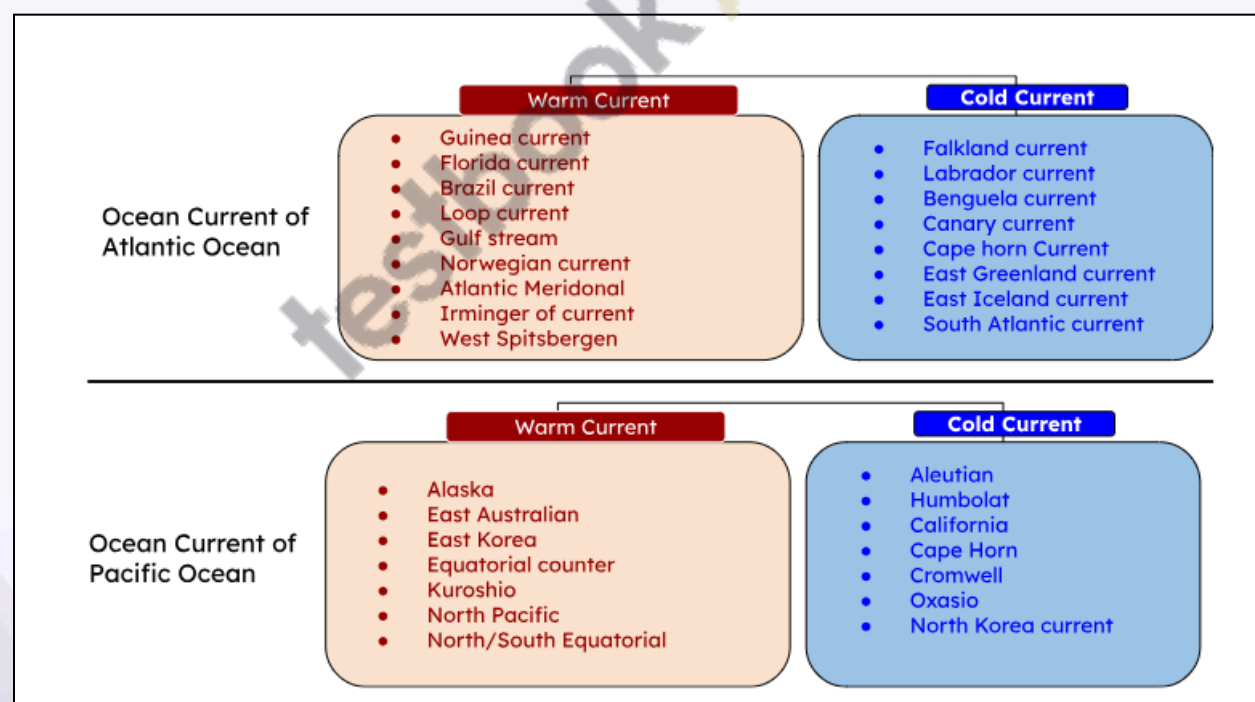
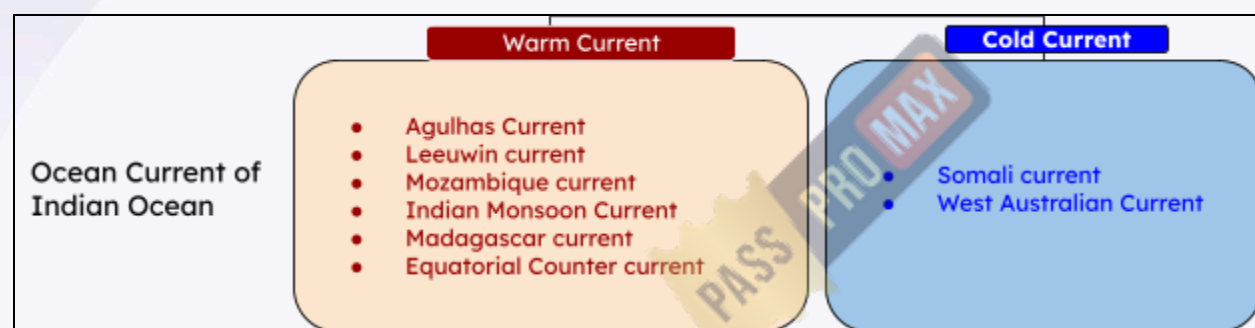
Important Ocean Currents of the World

Ocean	Hot/Cold	Name of Current	Features
Atlantic Ocean	Warm	Gulf Stream	Warm, strong current from the Gulf of Mexico, affecting the North Atlantic climate, especially in Europe.
Atlantic Ocean	Cold	Labrador Current	Cold current along Newfoundland's coast, impacting the eastern coast of North America.
Atlantic Ocean	Warm	Brazil Current	Southward-flowing warm current along Brazil, raising temperatures in southeastern South America.
Atlantic Ocean	Cold	Canary Current	Cold current moving south along North Africa, contributing to the arid climate of the Sahara.
Atlantic Ocean	Warm	North Atlantic Drift	Warm continuation of the Gulf Stream, bringing mild conditions to Western Europe.
Atlantic Ocean	Cold	Benguela Current	Cold current flowing north along Africa's southwest coast, influencing the Namib Desert climate.
Atlantic Ocean	Warm	North Equatorial Current	Westward-flowing warm current near the equator, part of the Atlantic current system.

Atlantic Ocean	Warm	South Equatorial Current	Warm current flowing westward in the southern Atlantic, between Africa and South America.
Atlantic Ocean	Cold	Falkland Current	Cold current flowing northward off Argentina, meeting the warm Brazil Current.
Atlantic Ocean	Warm	Norwegian Current	Warm extension of the North Atlantic Drift, moderating Norway's coastal climate.
Pacific Ocean	Warm	Kuroshio Current	Northward-flowing warm current off Japan, impacting climate in East Asia.
Pacific Ocean	Cold	California Current	Cold current flowing south along California, bringing cool and dry conditions to the region.
Pacific Ocean	Warm	East Australian Current	Warm current flowing south along Australia's east coast, warming the region.
Pacific Ocean	Cold	Peru (Humboldt) Current	Cold current flowing north along South America's west coast, contributing to arid climates.
Pacific Ocean	Warm	North Equatorial Current	Westward-flowing current in the northern Pacific, warm, feeds into the Kuroshio Current.
Pacific Ocean	Cold	Oyashio Current	Cold current flowing south from the Arctic along Japan, rich in nutrients.
Pacific Ocean	Warm	South Equatorial Current	Warm current in the southern Pacific, flowing westward across the ocean.
Pacific Ocean	Cold	Alaskan Current	Cold current flowing southward in the Gulf of Alaska, bringing cooler conditions to western Alaska.

Pacific Ocean	Warm	Equatorial Counter Current	Flows eastward between North and South Equatorial Currents, counteracting their westward flows.
Pacific Ocean	Cold	Kamchatka Current	Extension of the Oyashio Current, cooling the eastern coast of Russia.
Indian Ocean	Warm	Agulhas Current	Warm current flowing south along Africa's east coast, contributing to the Indian Ocean gyre.
Indian Ocean	Cold	West Australian Current	Cold current moving north along Australia's west coast, impacting southwestern Australian climate.
Indian Ocean	Warm	Somali Current	Seasonal current along Somalia's coast, reversing direction with monsoon winds, aiding local fisheries.
Indian Ocean	Warm	Mozambique Current	Warm current between Madagascar and Africa, merging into the Agulhas Current.
Indian Ocean	Warm	South Equatorial Current	Warm current flowing westward, driving water toward the east coast of Africa.
Indian Ocean	Cold	Indian Monsoon Drift	Seasonal current changing direction with monsoons, flows clockwise in summer, counterclockwise in winter.
Southern Ocean	Cold	Antarctic Circumpolar Current	Encircles Antarctica, transporting large volumes of water and nutrients, crucial for global circulation.
Southern Ocean	Cold	East Wind Drift	Eastward flow close to the Antarctic coast, counterbalancing the Antarctic Circumpolar Current.

Arctic Ocean	Cold	East Greenland Current	Cold current flowing southward along Greenland's east coast, contributing to ice drift in the Arctic.
Arctic Ocean	Cold	Beaufort Gyre	Circulates clockwise in the western Arctic, trapping freshwater and impacting ice cover.
Arctic Ocean	Cold	Labrador Sea Current	Part of the Labrador Current, impacts the North Atlantic's climate, particularly in winter.



Ocean Currents of Southern Ocean

Cold Current

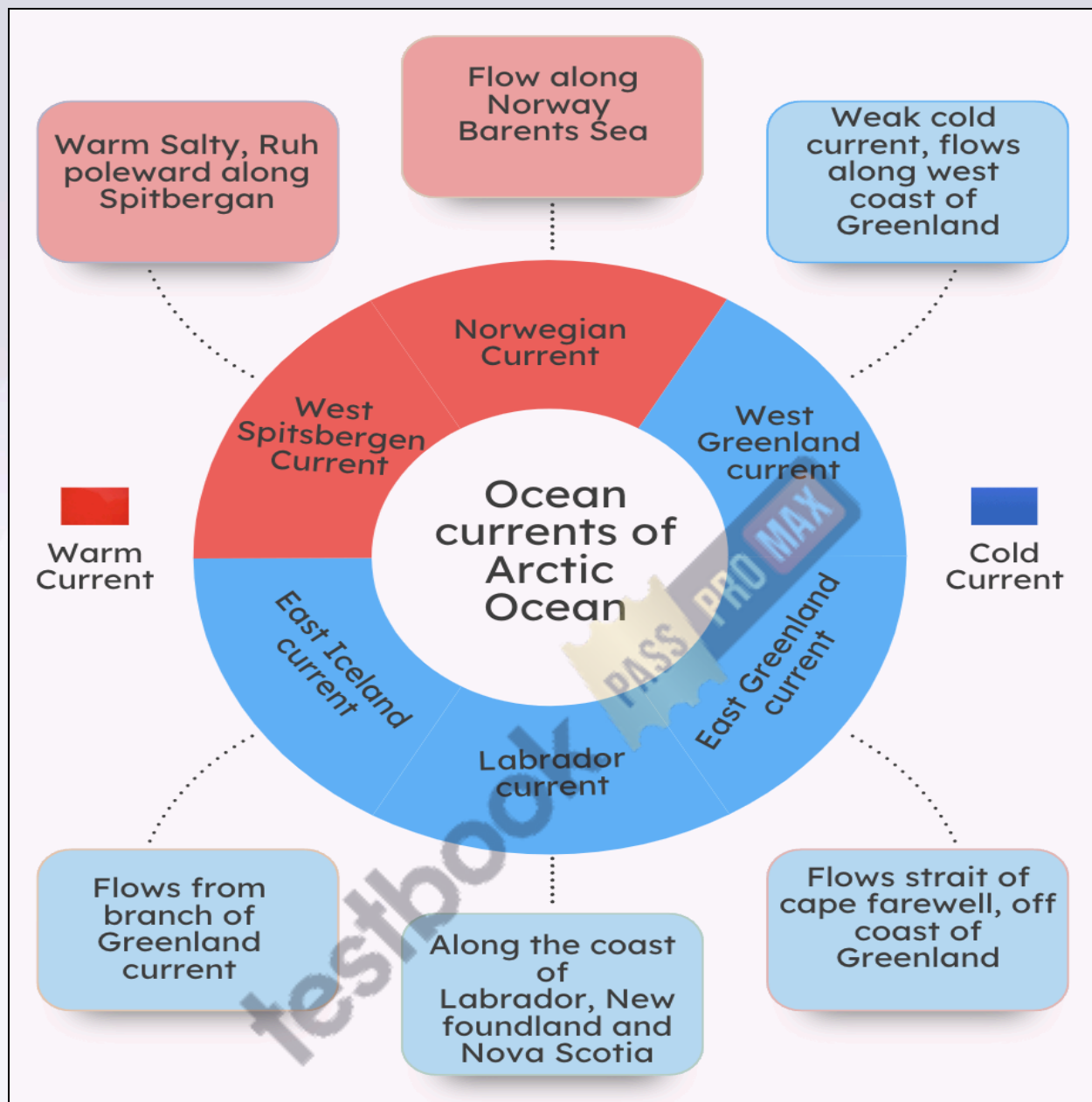
**Antarctic
Circumpolar
Current**

It flows clockwise
from west to east
around Antarctica

Cold current

**Tasman
Outflow**

Deep ocean current
flow from Pacific to
Indian Ocean



Effects of Ocean Currents

Ocean currents have a number of direct and indirect influences on human activities.

1. Climatic influence:

The west coasts of the continents in tropical and subtropical latitudes (except close to the equator) are bordered by cool waters. Their average temperatures are relatively low with narrow diurnal and annual ranges.

There is fog, but generally, the areas are arid. West coasts of the continents in the middle and higher latitudes are bordered by warm waters which cause a distinct marine climate. They are characterized by cool summers and relatively mild winters with a narrow annual range of temperatures.

Warm currents flow parallel to the east coasts of the continents in tropical and subtropical latitudes. This results in warm and rainy climates. These areas lie in the western margins of the subtropical anticyclones.

Rains:

- Warm ocean currents bring rain to coastal areas and even interiors. Example: Summer Rainfall in the British-type climate.
- Warm currents flow parallel to the east coasts of the continents in tropical and subtropical latitudes. This results in warm and rainy climates. These areas lie in the western margins of the subtropical anticyclones.

Drizzle:

- The mixing of cold and warm ocean currents creates foggy weather where precipitation occurs in the form of drizzle [Newfoundland].

Tropical cyclones:

- They pile up warm waters in the tropics and this warm water is the major force behind tropical cyclones.

2. Fishing Grounds:

The mixing of warm and cold currents helps to replenish the oxygen and favor the growth of plankton, the primary food for the fish population. The best fishing grounds in the world exist mainly in these mixing zones.

3. Desert Formation:

Cold ocean currents have a direct effect on **desert formation** in west coast regions of the **tropical and subtropical continents**. There is **fog** and most of the areas are **arid due to desiccating effect (loss of moisture)**.

The aridity of the hot deserts is mainly due to the effects of off-shore Trade Winds, hence they are also called **Trade Wind Deserts**.

- The major hot deserts of the world are located on the western coasts of continents between latitudes 15° and 30° N. and S.